# Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

## Lesson and revision-note context

| | |
|---|---|
| **Subject** | Mathematics |
| **Grade** | JSS1 |
| **Term** | Third Term |
| **Week** | 6 |
| **Topic** | Construction |
| **Subtopic** | Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees |
| **Theme** | — |
| **Curriculum objectives** | • construct parallel lines<br>• construct perpendicular lines<br>• bisect a given line segment<br>• construct angle 90°<br>• construct angle 60° |
| **Previous knowledge** | • Students have been taught how to draw lines. |
| **Lesson sequence** | • construct parallel lines — the teacher draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student construct parallel lines, following the teacher's board construction and replicating it in their notes<br>• construct perpendicular lines — the teacher draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student follow the teacher's board construction and replicate it in their notes<br>• bisect a given line segment — the teacher draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws; the student construct line segment, following the teacher's board construction and replicating it in their notes<br>• construct angle 90° — the teacher constructs angle 90 degrees on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs; the student construct angle 90 degrees, following the teacher's board construction and replicating it in their notes<br>• construct angle 60° — the teacher constructs angle 60 degrees on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs; the student construct angle 60 degrees, following the teacher's board construction and replicating it in their notes |
| **Instructional materials** | • Mathematical set. |
| **Evaluation** | — |

## Lesson objectives

| | |
|---|---|
| **1.** | construct parallel lines |
| **2.** | construct perpendicular lines |
| **3.** | bisect a given line segment |
| **4.** | construct angle $90^\circ$ |
| **5.** | construct angle $60^\circ$ |

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

# STUDENT REVISION NOTE

**Topic:** Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

**Class:** JSS1 **Subject:** Mathematics **Week:** 6

## Section 1: Lines That Never Meet

When you look at railway tracks stretching into the distance, the two rails never cross no matter how far they go. Lines that stay exactly the same distance apart and never meet are called parallel lines. They are written with the symbol ∥, so you would read AB ∥ CD as "line AB is parallel to line CD."

In this section you will learn to construct a line that is parallel to a given line and passes through a chosen point, using only a ruler and a pair of compasses. Constructing means drawing it accurately with these tools, not just sketching it by eye. This is a skill you will use again and again, because parallel lines appear all over geometry.

Here is the method for constructing a parallel line. Suppose you are given a line AB and a point P above it, and you want to draw the line through P that is parallel to AB.

Step 1. Draw the given line AB and mark the point P somewhere above it.

Step 2. Place the point of your compass on A and open it to the point P. Without changing this opening, swing an arc that crosses the line AB. Call the crossing point C.

Step 3. Now move the compass point to C and draw an arc above the line (still keeping the same opening) so that it passes through P. This arc marks out the path a circle would take through P at the same distance from the line.

Step 4. Without changing the compass opening, place the compass point on P and draw an arc that crosses the line AB. Call this new point D.

Step 5. Measure the distance from C to D with your compass. Then, keeping that same opening, put the compass point on P and draw an arc that crosses the first arc you drew through P. Call that crossing Q.

Step 6. Draw a straight line from P to Q with your ruler. This line is parallel to AB. You can check it: the arcs you drew guarantee that CP and DQ are equal in length, which is exactly what makes the two lines stay the same distance apart.

Worked example. Construct a line through the point P that is parallel to the given line AB.

Solution. Follow the six steps in order.

1. Draw line AB and mark point P above it.
2. Put the compass point on A, open it to P, and swing an arc down to cut the line at C.
3. Move the compass to C and, keeping the opening fixed, draw an arc upward through P.
4. Still keeping the same opening, put the compass point on P and draw an arc down to cut the line at D.
5. Measure the gap from C to D with the compass. Keep that opening, put the point on P, and draw an arc that crosses the arc from Step 3. Mark the crossing Q.
6. Join P and Q with a ruler.

The line PQ you have drawn is parallel to AB, so you can label it PQ ∥ AB. The key to the whole construction is that you kept the compass opening the same each time, which copies distances exactly onto your page.

**Practice**

1. What symbol do we use to show that two lines are parallel? Write it down and use it to write "line XY is parallel to line MN".

2. In the construction of a parallel line, you kept your compass opening exactly the same while drawing the arcs. Why is keeping the compass opening fixed so important in this construction?
3. Draw a line and label its ends A and B. Mark a point P above the line. Then construct the line through P that is parallel to AB, using a ruler and compasses. Describe each step you take in the order you do them.
4. Two lines are drawn so that they never meet, no matter how far they are extended, and the distance between them stays the same all the way along. What are these two lines called?

## Section 2: Constructing Perpendicular Lines

Two lines are **perpendicular** when they cross at a right angle — a square corner of exactly 90°. You meet perpendicular lines all the time: the floor meets the wall, the page edge meets its top edge, the rungs of a ladder meet the side rails.

You can construct a perpendicular line with just a **ruler** and a **compass** — no measuring of angles needed. The trick is that the compass draws equal distances, and equal distances force the corner to come out exactly right.

Here is the construction to draw a line perpendicular to a given line through a point on it.

**Step 1.** Draw a line and mark a point P on it where you want the perpendicular to stand.

**Step 2.** Put the compass point on P and open it a little. Draw an arc that crosses the line on both sides of P. Where that arc cuts the line, mark the two points A and B. Because you did not move the compass between the two cuts, PA and PB are exactly equal.

**Step 3.** Open the compass a little wider. Put the point on A and draw a small arc above the line. Without changing the compass opening, put the point on B and draw another small arc. The two arcs meet at a point C above the line.

**Step 4.** Join P to C with your ruler. The line PC is perpendicular to the original line.

Why does this work? In Step 3 the point C is exactly the same distance from A as from B. So C is perched perfectly above the middle of the segment AB, and the line from P to C stands straight up from the line — meeting it at exactly 90°.

Sometimes you need a perpendicular that comes down **onto** the line from a point above it. This is the same idea, flipped around.

**Step 1.** Draw a line and mark a point P **above** the line, off it.

**Step 2.** Put the compass point on P and open it wide enough that the arc will reach the line. Draw an arc that cuts the line in two places. Mark those two cuts A and B.

**Step 3.** Put the compass point on A and draw a small arc below the line. Without changing the compass opening, put the point on B and draw another small arc. The two arcs meet at a point C below the line.

**Step 4.** Join P to C. This line crosses the line at a right angle — it is perpendicular to the line and it passes through P.

Whether the point P sits on the line or above it, the method is the same: mark two equal distances from the line, find the point where the two wider arcs cross, and join. The crossing point is always exactly above the middle of your two marks, which forces the perpendicular.

A good habit before you finish any construction is to look at the corner where the two lines meet. If you have been careful, the corner should look like a clean square corner — and when you check later, it will measure exactly 90°.

**Example: construct a perpendicular to line XY at the point P.**

We follow the four steps with the line already drawn and P marked on it.

1. Put the compass point on P and draw an arc that cuts the line at A on the left and B on the right. The compass is not moved between the two cuts, so PA = PB.
2. Widen the compass a little. Draw an arc from A, then an arc from B with the same opening. They cross at C above the line.
3. Use the ruler to join P to C.

The line PC now stands on the line AB at a square corner. We have constructed a perpendicular line at P. If we check the corner with a set square or later construct a 90° angle there, it comes out exactly right because C was found at equal distances from A and B, putting it directly above the middle of AB.

**Practice**

1. Draw a horizontal line and mark a point P on it. Using only a ruler and compass, construct the line perpendicular to your line through P. In your working, say what you did in each of the four steps.
2. Draw a line and mark a point P above the line, off it. Using only a ruler and compass, construct the perpendicular from P down onto the line. Describe each step you took.

## Section 3: Finding the Midpoint: Bisecting a Line Segment

When we **bisect** a line segment, we cut it into two equal halves. The point where the two halves meet is called the **midpoint** — it sits exactly halfway along the segment. Bisecting a segment also gives us a new line that meets the segment at a right angle (90°), so the line we draw is called the **perpendicular bisector**.

To bisect a line segment you need only a ruler and a pair of compasses. Follow these steps:

1. Draw the line segment AB.
2. Open your compass to **more than half** the length of AB.
3. Place the compass point on A and draw an arc that sweeps **above and below** the segment.
4. Without changing the compass opening, place the point on B and draw arcs that sweep above and below, crossing the first two arcs.
5. Use your ruler to join the two points where the arcs cross. This new line is the **perpendicular bisector**. It crosses AB at the midpoint M, and it does so at a right angle, so AM = MB.

Why must the compass be opened to more than half the length? If the opening were less than half, the arcs drawn from A and from B would not reach each other, and there would be no crossing points to join. The two crossing points are the key: the line drawn through them is exactly what splits AB into two equal parts.

**Example: Bisect a line segment 6 cm long.**

1. Draw a line segment AB that is 6 cm long.
2. Open the compass to 4 cm (more than half of 6 cm).
3. With the compass point on A, draw an arc that sweeps above and below the segment.
4. Keeping the same opening, put the point on B and draw arcs above and below that cross the first two.
5. Join the two crossing points with a straight line. This is the perpendicular bisector.

The perpendicular bisector crosses AB at point M. Since it cuts AB into two equal parts, AM = MB = 6 ÷ 2 = 3 cm. Each half is 3 cm long, and the bisector meets AB at a right angle (90°).

**Practice**

1. Draw a line segment 8 cm long and bisect it with compasses. How long is each half after you bisect it?
2. When you bisect a line segment, the line you draw crosses the segment at what kind of angle, and what is that point called?
3. You are bisecting a segment AB that is 10 cm long. You open your compass to only 3 cm before drawing the arcs. Explain why your construction will fail, and state the smallest opening that would work.
4. You bisect a segment and find that one of the two halves is 7 cm long. What was the total length of the original segment?

## Section 4: Building a Perfect Right Angle: Constructing 90°

A right angle measures exactly 90°, and you can build one with only a ruler and a compass. The secret is to think of a straight line as a flat angle of 180°. If you split a straight line exactly in half at a point, each half is 90°.

Here is the construction that does exactly that. You will need a pencil, a ruler, and a compass.

Constructing a 90° angle at a point O on a line

Step 1 – Draw the baseline. Use your ruler to draw a straight line. Choose a point O on that line. This is where the 90° angle will sit.

Step 2 – Make two marks at equal distances. Put the point of your compass at O. Set the opening to any comfortable width and draw an arc that crosses the line. The arc cuts the line at two points, A on the left of O and B on the right of O. Because one arc drew them both, OA and OB are the same length.

Step 3 – Widen the compass. Open the compass so its width is bigger than OA (more than half of AB). This wider opening is important – it is what makes the two arcs you are about to draw cross each other.

Step 4 – Draw two arcs that meet. Keep the compass at this wider opening. Put the point at A and draw a gentle arc above O. Then, without changing the opening, put the point at B and draw another arc. The two arcs cross at a point above O – call it P.

Step 5 – Join the top. Use your ruler to draw a straight line from O through P. The line OP stands straight up from the baseline, so the angle between OP and the line (angle POA) is exactly 90°.

The key idea: the two equal arcs from A and B meet at P because P is the same distance from A as it is from B, and O is the midpoint of AB. So P is exactly halfway around the flat line from A to B – and halfway around 180° is 90°.

Worked example: Construct a 90° angle at point O.

Here is the full construction, one step at a time, so you can see exactly what your drawing should look like as you go.

1. Draw a horizontal line with your ruler. Mark a point O somewhere on it.
2. With the compass point at O, draw an arc crossing the line on both sides. It touches the line at A (left) and B (right). OA = OB.
3. Widen the compass so the opening is clearly larger than OA.
4. Keeping that wider opening, put the point at A and draw an arc above the line. Without changing the opening, put the point at B and draw a second arc. The two arcs cross above O – mark the crossing P.
5. Draw a straight line from O through P with your ruler.

The finished drawing has a vertical line OP standing on the horizontal line, with the corner at O. The angle between the baseline and OP, measured on either side, is 90°. Check it with the corner of your set square: it fits exactly.

**Practice**

1. Describe, in the right order, the five steps you take to construct a 90° angle at a point O on a straight line. (You may use a ruler and compass.)
2. In the 90° construction, you draw a first arc from O that cuts the line at A and B, then you draw two further arcs from A and B that meet at P. Why must the two arcs you draw from A and from B use exactly the same compass opening?
3. After constructing your 90° angle, you measure the other angle on the other side of OP, between the baseline and OP but on the opposite side. How big is that angle, and why?

## Section 5: Drawing a 60° Angle with a Compass

An angle of 60° is special because it is one of the angles you can build using only a ruler and a compass — no protractor needed. It turns out that if you draw an equilateral triangle (a triangle where all three sides are equal), then all three of its angles measure 60°. So the trick is to use the compass to mark a point that is the same distance from two points on your line, and the lines joining them will meet at 60°.

Here is the method, step by step. You will need a sharp pencil, a ruler, and a compass.

Step 1: Draw a straight line and pick a point on it. Let us call the line $\ell$ and the point A. The angle you build will have its corner (its vertex) at A.

Step 2: Put the point of the compass at A and open it to any convenient width. Draw an arc that crosses the line $\ell$. Call the point where the arc meets the line B. Keep the compass at exactly this same width for the next step — do not change it.

Step 3: Keeping the compass at that same width, put its point on B and draw an arc that passes through A and goes up above the line. Where this new arc crosses your first arc, mark that point C.

Step 4: Use the ruler to draw a straight line from A through C and extend it. The angle between your original line $\ell$ and this new line AC is 60°.

Why does this work? The compass keeps the distance AB the same everywhere, so AB = AC = BC. That makes triangle ABC an equilateral triangle, and every angle in an equilateral triangle is 60°. The angle at A between lines $\ell$ and AC is therefore 60°.

Example: Construct a 60° angle at point P on a line.

1. Draw a line and mark a point P on it. This line will be the bottom side of your angle.
2. Put the compass point on P and open it to a comfortable width (about halfway across your paper is fine). Draw an arc crossing the line to the right of P. Where the arc meets the line, call it Q.
3. Do not change the compass width. Put the point on Q and draw an arc that passes over P and rises up above the line. Where this new arc crosses the first arc, call it R.
4. Use the ruler to draw a straight line from P through R, extending it beyond R.
5. The angle between your original line PQ and the new line PR is 60°.

Check your work: PR is a radius of the first arc, so it equals PQ. And RQ is a radius of the second arc, so it also equals PQ. That means PQ = PR = RQ, so triangle PQR has three equal sides — it is equilateral, and every angle inside it is 60°. The angle you drew at P is the 60° one.

**Practice**

1. You are constructing a 60° angle at point M on a straight line. After drawing the first arc that crosses the line at point N, you must keep the compass at the same width. In the next step you put the compass point on N. How far should the compass be opened, and what single property of the figure does keeping it this width guarantee?
2. A classmate says: "To construct a 60° angle, I draw the first arc, then I can use any compass width for the remaining arcs — it doesn't matter if I change it." Explain whether this is correct, and say what happens to the angle if the compass width is changed partway through.

## Check your work

Attempt every question before you read this page.

### Lines That Never Meet

1. The parallel symbol is $\parallel$, and "line XY is parallel to line MN" is written XY $\parallel$ MN.
2. Keeping the compass opening fixed copies the same distance onto the page each time, so the two segments that control the spacing are equal. Equal spacing is what makes the two lines stay the same distance apart and therefore parallel.
3. Steps: (1) Place the compass point on A, open to P, and swing an arc down to cut AB at C. (2) Put the point on C and draw an arc up through P. (3) Put the point on P and draw an arc down to cut AB at D. (4) Measure CD with the compass, then put the point on P and draw an arc crossing the arc from step 2 at Q. (5) Draw PQ with the ruler. PQ is parallel to AB.
4. They are parallel lines.

### Constructing Perpendicular Lines

1. Draw the line and mark P. With the compass point on P, draw an arc cutting the line at A (left) and B (right). Widen the compass and draw an arc from A and an arc from B with the same opening; mark where they cross as C. Join P to C with the ruler. PC is the perpendicular at P.

   Step 1: draw the line and mark P. Step 2: compass on P, draw an arc that cuts the line at A and B, keeping the opening fixed so PA = PB. Step 3: widen the compass, draw arcs from A and from B with the same opening, meeting at C above the line. Step 4: join P to C. Because C is equally distant from A and B, PC meets the line at exactly 90°, so PC is perpendicular.

2. Put the compass point on P and draw an arc wide enough to cut the line at A and B. Then, keeping the opening fixed, draw an arc from A and an arc from B below the line; mark where they cross as C. Join P to C. The line PC crosses the original line at a right angle and passes through P, so it is the required perpendicular.

   With the compass on P, draw an arc reaching the line and cutting it at A and B (P is equally distant from both cuts). Without changing the opening, draw an arc from A and an arc from B below the line; they meet at C. Joining P to C gives a line through P that meets the original line at exactly 90°, since C lies directly below the middle of AB.

### Finding the Midpoint: Bisecting a Line Segment

1. Each half is 4 cm long.

   The perpendicular bisector cuts the segment into two equal parts. Half of 8 cm is $8 \div 2 = 4$ cm, so AM = MB = 4 cm.
2. The line crosses the segment at a right angle (90°), and the point where it crosses is the midpoint.
3. 3 cm is less than half of 10 cm, so the arcs drawn from A and from B will not reach each other and there will be no crossing points to join. The smallest opening that works is anything more than 5 cm (more than half of 10 cm).

   For the arcs to cross, the compass opening must be more than half the length of the segment. Half of 10 cm is 5 cm, so the opening must be greater than 5 cm, for example 6 cm. With 3 cm the arcs never meet, so no crossing points exist and the bisector cannot be drawn.
4. 14 cm.

   Bisecting splits the segment into two equal halves. If one half is 7 cm, the other half is also 7 cm. Total length = 7 cm + 7 cm = 14 cm.

### Building a Perfect Right Angle: Constructing 90°

1. a. Draw a straight line and mark point O on it.
   b. With the compass point at O, draw an arc crossing the line at A (left) and B (right), so OA = OB.
   c. Widen the compass so its opening is larger than OA.
   d. With the wider opening, draw an arc from A above O, then without changing the opening draw an arc from B; the arcs cross at P.
   e. Draw the straight line from O through P. Angle POA = 90°.

   The construction builds a 90° angle by halving the flat 180° line. Draw the baseline and pick O. One arc from O marks equal distances A and B. A wider compass then draws two arcs from A and B that meet at P. Joining O to P splits the straight line exactly in half, so the angle between OP and the baseline is 90°.
2. They must use the same opening so that AP and BP are equal in length. That makes P the same distance from A as from B, so the line OP passes exactly through the middle of the flat 180° line, giving 90°.

   An arc drawn with the compass point at A and a fixed opening reaches every point the same distance AP from A. A second arc with the same opening from B reaches every point the same distance BP from B. Where the two arcs cross, AP = BP. So P is balanced between A and B, and joining O (the midpoint) to P cuts the straight angle in half to give 90°.
3. It is also 90°. The whole line is a straight angle of 180°, and OP divides it into two equal angles, so each is 90°.

   A straight line forms a flat angle of 180°. The perpendicular line OP splits it into two equal angles on either side. Half of 180° is 90°, so both the angle on one side and the angle on the other side are 90°.

### Drawing a 60° Angle with a Compass

1. The compass must stay open to the same width as when you drew the first arc (the distance from M to N). Keeping it the same width guarantees that the three sides of the triangle you build are equal, so the triangle is equilateral and every angle inside it measures 60°. The angle you construct at M is therefore 60°.

   When you set the compass at M and mark N, the compass width equals the distance MN. If you never change this width, then the next point you mark (call it P) is also at distance MN from both M and N. So

MN = NP = PM, and triangle MNP has three equal sides. An equilateral triangle has three equal angles that add to 180°, so each is 180° ÷ 3 = 60°.

2. It is not correct. To get exactly 60° you must keep the compass at the same width for every arc. If you change the width partway through, the three marked points will not all be the same distance from each other, so the triangle will not be equilateral. The angle you draw will no longer be exactly 60° — it could be smaller or larger, but it will not be 60°.

   The 60° result comes entirely from the triangle having three equal sides. If every arc uses the same width, then each side of the triangle equals that width, so the triangle is equilateral and all its angles are 60°. If the width changes, one or more sides have different lengths, the triangle is no longer equilateral, and the constructed angle is not 60°. Keeping the width fixed is what makes the construction reliable.

## Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

| | | | |
|---|---|---|---|
| **Revision note ID** | cand-bcada69c | **Date** | |
| **Subtopic** | Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees | **Rater** | |

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

| **Criterion** (select one score per row) | **1** Poor | **2** Weak | **3** Adequate | **4** Good | **5** Excellent |
|---|---|---|---|---|---|
| **Objective and curriculum alignment** Does the revision note cover the intended lesson objectives and curriculum target? | ○ | ○ | ○ | ○ | ○ |
| **Taught-lesson alignment** Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth? | ○ | ○ | ○ | ○ | ○ |
| **Scope and conceptual correctness** Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors? | ○ | ○ | ○ | ○ | ○ |
| **Student-facing clarity and usefulness** Can JSS1 learners understand and use it meaningfully as take-home review material? | ○ | ○ | ○ | ○ | ○ |

| Strengths | Concerns | Recommendation |
|---|---|---|
| | | |

## Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

| | |
|---|---|
| ☐ | **Misses a required objective** — Skips something the lesson was meant to teach. |
| ☐ | **Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic). |
| ☐ | **Wrong or misleading content** — Contains a mathematical or factual mistake. |
| ☐ | **Too hard for JSS1 to follow** — Likely confusing for the students. |
| ☐ | **Other:** _____ |